

Ryan Cho

Software Engineer — Full-Stack Web Applications

+1 608 360 1459 · Ryanmcho100@gmail.com · <https://www.linkedin.com/in/ryan-cho-2020/> · <https://github.com/Chomama> · New York, United States

PROFESSIONAL SUMMARY

Full-stack software engineer specializing in React and TypeScript single-page applications, GraphQL and REST APIs, and scalable data integrations across regulated financial services. Delivers performant multi-product frontends, instruments services for metrics and alerting on AWS, and partners with product and engineering leaders to ship reliable features. Strong foundation in algorithms, database design, and containerized cloud deployment.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led **frontend architecture evolution** for a multi-product React and TypeScript suite, establishing shared component patterns and Webpack optimizations that improved initial load times across portfolio analytics workflows.
- Designed and shipped **GraphQL and REST API extensions** powering new trading, reporting, and dashboard features while maintaining backward compatibility for downstream consumers.
- Instrumented the **API layer for metrics collection, monitoring, and alerting**, integrating distributed tracing and SLA dashboards that shortened mean time to detect production regressions.
- Built pipelines integrating **operational data capture outputs** into a descriptive analytics platform, enabling portfolio teams to query normalized datasets for ad hoc analysis.
- Partnered with product and engineering leadership to deliver a **greenfield analytics module** from inception through production release, aligning UX with complex investment domain workflows.
- Refactored legacy JavaScript modules into **typed TypeScript services**, reducing defect rates in high-traffic code paths and simplifying onboarding for new engineers.
- Deployed containerized microservices on **AWS with Docker and Kubernetes**, coordinating zero-downtime rollouts and environment parity across staging and production.
- Mentored peers through **design and code reviews**, raising frontend quality standards and advocating for simpler, maintainable solutions to cross-team integration problems.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **React single-page applications** for risk and trading operations, translating complex regulatory workflows into intuitive interfaces used daily by front-office analysts.
- Built and maintained **REST microservices** in Node.js and Java, exposing portfolio and reference data to internal consumers with consistent pagination and error contracts.
- Modeled and queried **PostgreSQL and MongoDB** schemas supporting trade lifecycle reporting, balancing relational integrity with flexible document storage for semi-structured feeds.
- Contributed to **API observability** by adding structured logging, latency metrics, and alert thresholds that improved visibility into batch and real-time processing jobs.
- Optimized existing **JavaScript rendering paths** and database queries, cutting dashboard refresh latency for high-volume intraday monitoring screens.
- Collaborated with cross-functional stakeholders to define **software design specifications** for regulatory reporting tools, incorporating feedback from compliance and operations teams.
- Supported **AWS-hosted deployments** and Docker-based local environments, helping standardize build pipelines across multiple application teams.

Software Engineering Intern Chexology

Jun/2018 - May/2019

Remote | 17th Floor, New York

- Implemented **React components** for a restaurant SaaS platform, improving order management screens used by staff across multiple client locations.
- Extended **Node.js REST endpoints** supporting menu, inventory, and checkout flows with input validation and consistent JSON response shapes.
- Configured **Webpack build tooling** for staging releases, enabling faster iteration on frontend features without disrupting production deployments.
- Integrated **PostgreSQL-backed reporting queries** into admin dashboards, giving operators visibility into daily transaction volumes.
- Participated in **peer code and design reviews**, providing constructive feedback on UI patterns and backend error handling.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web pages** for a nonprofit fundraising site, ensuring accessible layouts across desktop and mobile viewports.
- Developed **JavaScript form validation and interactive UI** flows that improved donor completion rates on campaign landing pages.
- Collaborated with a remote team to publish **CMS-driven content updates**, maintaining consistent branding and navigation structure.
- Improved **page load performance** through image optimization and asset bundling, reducing time-to-interactive on content-heavy pages.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Webpack, SPA architecture, responsive UI, Chart.js, D3.js

Backend: Node.js, GraphQL, REST APIs, Express, Java, Python, API design, microservices

Databases: PostgreSQL, Redis, MongoDB, SQL modeling, query optimization, document and key-value stores

Cloud & DevOps: AWS (EC2, S3, Lambda, ECS), Docker, Kubernetes, CI/CD, infrastructure as code

Monitoring & Tools: Metrics instrumentation, alerting, distributed tracing, Datadog, Prometheus, Grafana, Git, logging pipelines

Workflow & PM: Agile/Scrum, Jira, technical design documentation, cross-functional collaboration, code review leadership